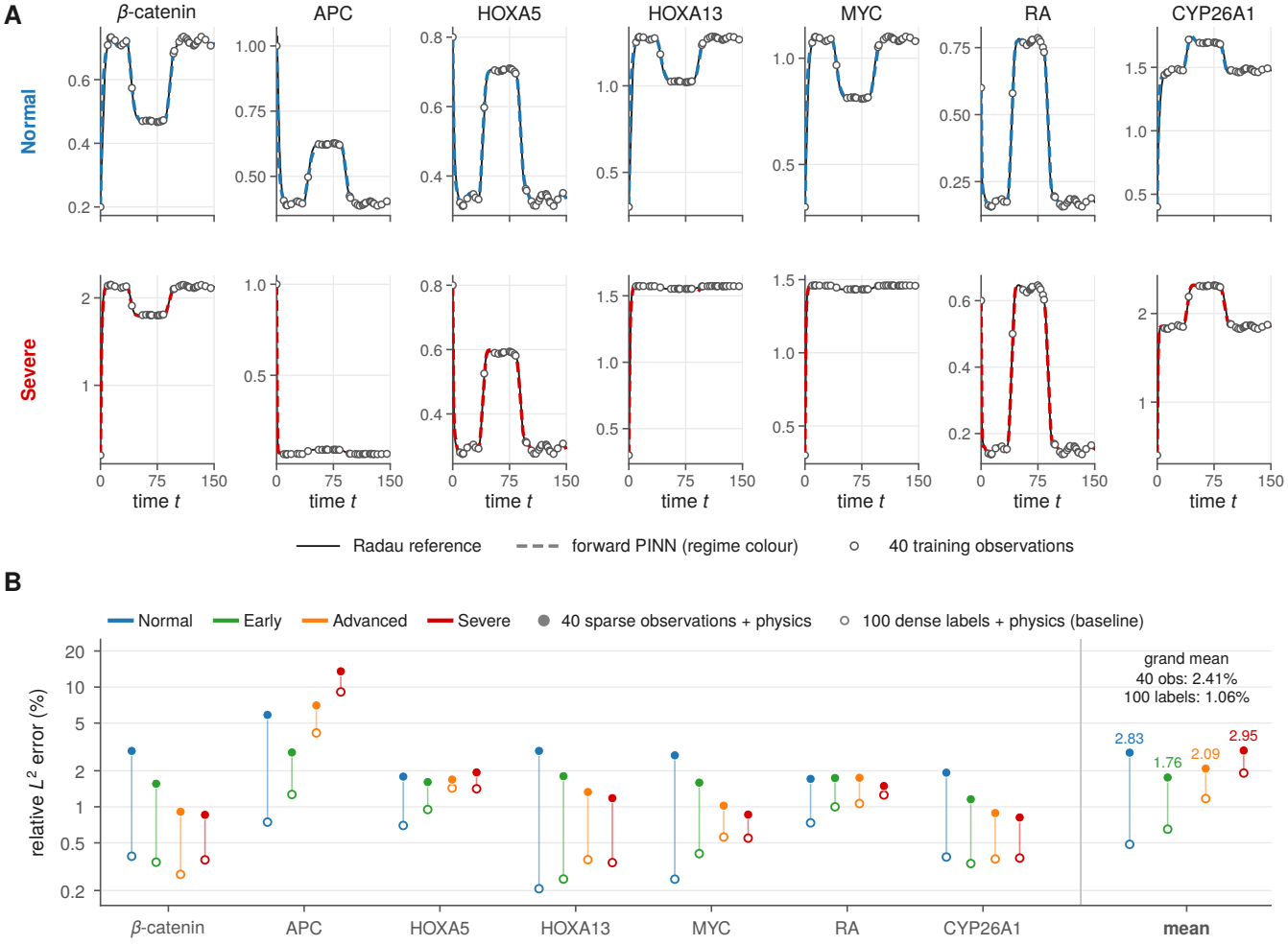

A
β-catenin
APC
HOXA5
HOXA13
MYC
RA
CYP26A1
Normal
Severe
time t
Radau reference
forward PINN (regime colour)
40 training observations
B
Normal
Early
Advanced
Severe
40 sparse observations + physics
100 dense labels + physics (baseline)
relative L² error (%)
grand mean
40 obs: 2.41%
100 labels: 1.06%
2.83
1.76
2.09
2.95
β-catenin
APC
HOXA5
HOXA13
MYC
RA
CYP26A1
mean